

Meta-Labeling with Provable Precision Guarantees for Safety-Critical Transportation Systems

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Highlights

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- Mathematical framework provides instance-level confidence with provable precision guarantees
- Entropy dominates probability magnitude for confidence assessment by factor of 2.7
- Achieves 96.9 percent automation coverage at 100 percent precision on 3.5 million transfers
- Framework generalizes to safety-critical transportation problems requiring human oversight
- Statistical validation includes McNemar testing and corrected cross-validation procedures

Meta-Labeling with Provable Precision Guarantees for Safety-Critical Transportation Systems

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Abstract

Transportation agencies increasingly deploy machine learning models for operational decisions affecting millions of daily trips, yet lack formal methods to identify which predictions warrant automation versus human review. Standard supervised learning produces point predictions or class probabilities without instance-level confidence assessment, forcing agencies to choose between full automation that risks deploying unreliable predictions or complete manual review that negates efficiency gains. This research develops a mathematical framework for trustworthy AI deployment in safety-critical transportation applications through meta-labeling: a two-stage learning architecture where base models predict outcomes and meta-models assess prediction reliability. The framework provides provable precision guarantees, entropy-based confidence scores, and formal methods for human-in-the-loop automation. Theoretical analysis demonstrates that prediction entropy dominates probability magnitude for confidence assessment because entropy quantifies model uncertainty regardless of prediction direction. Statistical validation employs McNemar testing for paired binary classifications and corrected cross-validation for confidence intervals. Empirical validation on 3.5 million scheduled transit transfers demonstrates 96.9 percent automation coverage at 100 percent precision using only static schedule data, with entropy coefficient magnitude 2.7 times larger than probability magnitude. The framework generalizes to transportation classification tasks requiring deployment confidence: pavement condition assessment, bridge structural analysis, traffic signal configuration, and similar applications where high-stakes decisions at scale necessitate both automation efficiency and human oversight.

Keywords: trustworthy artificial intelligence, meta-learning, confidence quantification, transit reliability, statistical validation, human-in-the-loop systems, mathematical foundations

1. Introduction

Transportation systems impose unique demands on artificial intelligence deployment. Safety-critical operations, interconnected infrastructure, and decisions affecting millions of daily trips create conditions where prediction errors carry asymmetric consequences. Transit agencies modifying schedules, traffic engineers re-configuring signals, and infrastructure managers prioritizing inspections require more than aggregate model accuracy. When a classification model achieves 98 percent accuracy on 15,000 predictions, the remaining 2 percent represents 300 errors. Without knowledge of which 300, practitioners face a binary choice: accept all predictions and risk deploying incorrect decisions, or review all predictions manually and eliminate efficiency gains. Standard machine learning provides no principled method to distinguish reliable predictions from uncertain ones at the instance level.

The challenge extends beyond accuracy to trustworthiness. Rajkomar et al. (2018) demonstrated that clinical decision support systems achieving 92 percent accuracy encountered resistance until explainable confidence scores identified uncertain predictions for review. Dal Pozzolo et al. (2018) showed that fraud

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detection systems requiring human approval for uncertain cases reduced false positives 67 percent while maintaining detection rates. Transportation applications face similar requirements: automated decisions must be accountable, transparent in their uncertainty, and integrated with human oversight where algorithmic certainty is low. Calibration methods ensure predicted probabilities align with empirical frequencies across populations (Guo et al., 2017), but do not identify which specific instances are uncertain. Conformal prediction provides distribution-free coverage guarantees (Vovk et al., 2005; Romano et al., 2019), yet produces set-valued predictions rather than scalar confidence scores suitable for threshold-based workflow optimization. Bayesian approaches model full posterior distributions (MacKay, 1992; Gal & Ghahramani, 2016), but computational costs prohibit batch processing millions of transportation decisions, and uncertainties quantify parameter estimates rather than prediction reliability.

This research develops mathematical foundations for trustworthy AI deployment through meta-labeling: a framework where base models predict outcomes and meta-models assess base prediction reliability. Adapted from quantitative finance (Lopez de Prado, 2018), meta-labeling constructs binary meta-labels indicating whether base predictions matched true outcomes in historical data, then trains meta-models to predict these labels using features derived from base predictions, input characteristics, and contextual information. The framework provides three theoretical contributions. First, it establishes formal conditions under which instance-level confidence assessment achieves provable precision guarantees, enabling agencies to specify precision requirements and derive coverage analytically through precision-recall optimization. Second, it demonstrates mathematically why prediction entropy dominates probability magnitude for confidence assessment: entropy quantifies model uncertainty about outcomes regardless of prediction direction, while probability magnitude quantifies only predicted likelihood. Third, the two-stage architecture provides natural interpretability through separation of outcome prediction and reliability assessment, addressing accountability requirements in safety-critical applications.

Empirical validation applies the framework to scheduled transit transfer reliability using 3.5 million transfers from Edmonton Transit Service. An XGBoost base model achieves perfect discrimination (ROC-AUC 1.0000, 2 errors in 497,522 test instances) on clearly labeled transfers derived from static General Transit Feed Specification data. A logistic regression meta-model predicts base correctness using 14 meta-features including prediction entropy, confidence scores, and schedule attributes. At confidence threshold 0.50, the system automates 96.9 percent of decisions at 100 percent precision, flagging 3.1 percent for manual review while capturing both base errors. Meta-feature analysis reveals entropy coefficient magnitude of negative 77.92 compared to probability magnitude of 28.90, a factor of 2.7 difference consistent with theoretical predictions. Statistical validation through McNemar testing confirms significant improvements over baseline models ($p < 0.001$), and 5-fold cross-validation produces zero variance in ROC-AUC across folds, demonstrating stable function learning rather than overfitting.

The framework addresses four critical gaps in transportation AI deployment. First, it provides instance-level confidence assessment where existing methods offer only population-level calibration or computationally prohibitive Bayesian inference. Second, it operates on static schedule data accessible to resource-constrained agencies lacking extensive automatic vehicle location archives, traffic sensors, or weather integration. Third, it delivers provable precision guarantees through threshold-based filtering, enabling risk-aligned deployment where agencies specify acceptable error rates and the system maximizes automation coverage subject to those constraints. Fourth, it generalizes beyond transit transfers to any transportation classification requiring deployment confidence: pavement condition assessment flagging uncertain predictions for field inspection, bridge structural analysis identifying edge cases for engineering review, traffic signal timing flagging atypical configurations for manual analysis.

This research contributes to mathematical foundations for trustworthy AI in transportation through five advances. First, it adapts meta-labeling theory from financial machine learning to transportation operations, formalizing the framework for safety-critical infrastructure decisions. Second, it provides information-theoretic justification for entropy dominance in confidence assessment, demonstrating that uncertainty measures outweigh likelihood measures for predicting reliability. Third, it develops statistical validation procedures for binary transportation classification including paired testing via McNemar's chi-squared and corrected resampled t-tests for cross-validation confidence intervals. Fourth, it establishes empirical validation on 3.5 million scheduled transfers demonstrating the framework achieves stated theoretical guarantees.

Fifth, it provides deployment guidelines for human-in-the-loop automation through precision-recall optimization aligned with operational constraints.

Section 2 reviews literature on transfer reliability prediction, uncertainty quantification, meta-learning, and operational deployment challenges. Section 3 formalizes the meta-labeling framework including triple-barrier label construction, entropy-based confidence features, and statistical validation procedures. Section 4 presents empirical validation results on base model performance, meta-labeling effectiveness, meta-feature analysis, and regime heterogeneity. Section 5 discusses theoretical foundations, generalizability to transportation problems, deployment considerations, and limitations. Section 6 concludes with implications for trustworthy AI in transportation systems.

2. Literature Review

This review examines four interconnected research areas: the evolution of transfer reliability prediction methods, uncertainty quantification techniques in predictive modeling, meta-learning frameworks for confidence assessment, and operational deployment challenges in transit planning. These areas collectively define the current state of knowledge and identify gaps that motivate the present investigation.

2.1. Evolution of Transfer Reliability Prediction

Transfer reliability prediction has progressed through three distinct methodological phases over four decades. Early deterministic optimization assumed perfect schedule adherence. Probabilistic methods emerged with automatic vehicle location systems. Machine learning approaches now dominate recent research. Each phase addressed limitations of its predecessor while introducing new challenges.

Deterministic optimization treated transfer reliability as a scheduling problem. Ceder (1989) developed mathematical programming formulations for short-turn design, establishing buffer time as the primary control variable under zero-variance assumptions. Bookbinder & Désilets (1992) extended this to network-wide transfer optimization with transfer penalties in objective functions. Hall et al. (2001) incorporated holding strategies for real-time control but maintained deterministic arrival assumptions. Ceder (2007) synthesized these approaches into planning guidelines that acknowledged operational variability conceptually while providing deterministic buffer formulas calibrated to historical percentiles. Field studies demonstrated schedule adherence rates of 60 to 75 percent during peak periods (Cats, 2011), invalidating the deterministic framework for operational prediction.

Automatic vehicle location systems enabled probabilistic modeling of operational uncertainty. Guo & Wilson (2011) developed gamma-distributed running time models from historical AVL data, demonstrating that travel time distributions exhibited right skewness inconsistent with normal assumptions. Nesheli & Ceder (2016) established lognormal distributions as statistically superior to gamma or normal alternatives across multiple transit systems, with Kolmogorov-Smirnov goodness-of-fit improvements of 15 to 22 percent. Cats (2011) developed agent-based simulation integrating vehicle trajectories with passenger dynamics, capturing cascade effects where upstream delays propagate through transfer chains. Minor delays at high-volume hubs produced disproportionate passenger impacts through missed connections. Cats & Jenelius (2015) applied Bayesian networks to predict missed connections using upstream delays, weather conditions, traffic congestion, and time-of-day variables. The model achieved 78 percent accuracy on the Stockholm network, representing a 31 percent improvement over schedule-based predictions, though requiring extensive historical AVL data, real-time traffic feeds, and weather integration. Wang et al. (2015) introduced survival analysis frameworks treating successful transfers as time-to-event outcomes with censoring for passengers who abandon waiting. This perspective enabled more nuanced treatment of passenger behavior but required boarding and alighting data that most agencies do not collect systematically. Full network simulation required 12 to 48 hours for single-day analysis (Cats, 2011), prohibiting operational use for thousands of daily schedule modifications.

Machine learning methods predict transfer outcomes directly from historical data without explicit distributional assumptions. Petersen et al. (2019) benchmarked linear regression, random forests, and gradient boosting for bus arrival prediction across five Danish cities. Gradient boosting achieved mean absolute errors

of 1.8 minutes with 95 percent confidence intervals of 1.6 to 2.0 minutes, outperforming linear models by 34 percent and random forests by 12 percent. Feature importance analysis revealed that schedule adherence of immediately preceding trips explained 67 percent of variance. Deep learning architectures demonstrated strength in temporal pattern extraction. Liu et al. (2020) applied LSTM networks to passenger flow forecasting, capturing weekly seasonality, holiday effects, and event-driven surges that eluded traditional time series methods. Zhang et al. (2020) developed attention mechanisms to identify influential upstream route segments, showing that delays on specific critical links predicted missed transfers with 0.89 precision compared to 0.71 for models using only local features. Sun et al. (2021) combined convolutional layers for spatial dependencies with recurrent architectures for temporal dynamics, achieving 8.7 percent improvement over standalone methods in multi-step-ahead prediction.

Graph-based approaches exploited network topology explicitly. Ke et al. (2021) developed graph convolutional networks representing transit systems as directed graphs with routes as edges and transfer points as nodes. This architecture improved prediction accuracy 23 percent over gradient boosting, with ROC-AUC of 0.94 compared to 0.76, by propagating information through network structure. The approach captured how congestion in one corridor affects downstream transfer reliability but required complete network representation and GPU acceleration.

Transfer-specific applications remained less developed than general arrival prediction. Huang et al. (2019) applied random forests to San Francisco transfers, achieving 92 percent accuracy using scheduled buffer time, historical adherence rates, route load factors, and weather indicators. Wu et al. (2020) improved interpretability through XGBoost with SHAP values, quantifying feature contributions for individual predictions. Their analysis revealed substantial heterogeneity: buffer time dominated predictions for short-headway routes at 75 percent feature importance, while schedule adherence dominated long-headway routes at 58 percent importance. Berrebi et al. (2018) conducted comprehensive evaluation across six bus arrival prediction models in Pittsburgh, noting that even models achieving 85 to 90 percent accuracy provided no mechanism to identify which specific predictions were reliable. A planner facing 15,000 transfers with 92 percent model accuracy confronts 1,200 potential errors but lacks guidance on which predictions warrant manual review.

2.2. Uncertainty Quantification in Predictive Models

Prediction accuracy alone provides insufficient information for operational deployment when errors carry asymmetric consequences. Uncertainty quantification addresses this through three frameworks: calibration methods that align predicted probabilities with empirical frequencies, prediction intervals that provide coverage guarantees, and Bayesian approaches that model full posterior distributions.

Calibration ensures that among instances assigned predicted probability p , the empirical frequency of the positive class approximates p . Platt (1999) introduced sigmoid calibration for support vector machines, applying logistic regression to transform uncalibrated scores into probabilities. Zadrozny & Elkan (2002) developed isotonic regression and binning approaches, distinguishing calibration (alignment of probabilities with frequencies) from discrimination (separation of classes). A model can achieve high accuracy through perfect discrimination while remaining poorly calibrated if probability estimates systematically deviate from empirical rates. Guo et al. (2017) demonstrated that modern deep neural networks, despite achieving state-of-the-art accuracy, produce poorly calibrated probabilities. Networks trained with maximum likelihood exhibit overconfidence: predictions near 0.9 correspond to empirical success rates of 0.75 to 0.80. Temperature scaling, a single-parameter post-hoc method that divides logits by learned temperature before softmax, reduced expected calibration error by 60 to 75 percent across vision and language tasks. Nixon et al. (2019) developed comprehensive diagnostic frameworks including reliability diagrams, expected calibration error, and adaptive binning schemes that adjust bin widths to ensure sufficient sample sizes. Calibration methods improve aggregate reliability across all predictions at 0.9 confidence but do not identify which specific instances will fail.

Prediction intervals provide coverage guarantees. Khosravi et al. (2011) reviewed neural network interval construction via bootstrap aggregation, training ensembles on resampled data and using prediction variance to estimate intervals. Quan et al. (2014) applied quantile regression to estimate conditional quantiles directly, predicting 10th and 90th percentiles of the response distribution without parametric assumptions. Conformal

prediction offers distribution-free coverage guarantees. Vovk et al. (2005) established theoretical foundations: given training data and exchangeability assumptions, conformal methods produce prediction sets guaranteed to contain the true label with user-specified probability, regardless of underlying data distribution. Romano et al. (2019) extended these guarantees to time series through adaptive conformal inference, maintaining coverage under non-stationary distributions. Transportation applications include Wang et al. (2021) for taxi demand forecasting and Li et al. (2021) for traffic speed prediction, both achieving nominal 90 percent coverage with empirical coverage of 89.2 and 90.8 percent respectively, with interval widths 30 to 40 percent narrower than parametric alternatives. Conformal methods produce set-valued predictions rather than scalar confidence scores. Bootstrap methods require training hundreds of models; conformal prediction requires recomputing conformity scores for each new instance. For transit networks with millions of scheduled transfers updated quarterly, these costs accumulate prohibitively. Romano et al. (2019) reported inference times 50 to 100 times slower than point prediction.

Bayesian approaches model full posterior distributions over parameters. MacKay (1992) pioneered Bayesian neural networks using Laplace approximations to posterior distributions, providing uncertainty estimates through second-order Taylor expansions around maximum a posteriori parameters. Neal (1996) developed Markov chain Monte Carlo methods for posterior sampling, enabling exact Bayesian inference at prohibitive computational cost: sampling 10,000 posterior draws for a 100,000-parameter network required weeks on 1990s hardware. Gal & Ghahramani (2016) demonstrated that dropout training implicitly performs variational Bayesian inference, enabling uncertainty estimation through test-time dropout: running multiple forward passes with random dropout and measuring prediction variance. Lakshminarayanan et al. (2017) proposed deep ensembles as a non-Bayesian alternative that empirically outperforms variational approximations in out-of-distribution detection. Wilson & Izmailov (2020) noted fundamental challenges in scaling Bayesian deep learning. Full posterior inference requires integrating over parameter space, computationally infeasible for networks with millions of parameters. Variational approximations restrict posterior to tractable families such as factorized Gaussians, potentially misrepresenting true uncertainty. Ovadia et al. (2019) benchmarked thirteen uncertainty methods on distribution shift tasks, finding that approximate Bayesian methods substantially underestimate uncertainty when test data diverge from training distributions. Ensemble methods require training and storing multiple models, increasing memory and computation five to ten fold. Dropout inference requires 50 to 100 forward passes per prediction (Gal & Ghahramani, 2016), prohibitive for batch processing. Bayesian uncertainties quantify parameter uncertainty given training data, distinct from the operational question of whether this specific prediction is trustworthy.

2.3. Meta-Learning and Confidence Assessment

Meta-learning operates at a higher level of abstraction than traditional machine learning: rather than learning patterns in data, it learns patterns across learning tasks. Caruana (1997) demonstrated that jointly training models on related tasks improves generalization by sharing representations that capture common structure. Training neural networks simultaneously for object detection, segmentation, and depth estimation produced superior performance on each individual task compared to independent training, with error reductions of 15 to 30 percent. Finn et al. (2017) developed model-agnostic meta-learning, which optimizes for rapid adaptation: rather than learning parameters that perform well on training tasks, the method learns parameters that require minimal fine-tuning to perform well on new tasks. Hospedales et al. (2021) synthesized three decades of meta-learning research, identifying key principles: learning across task distributions rather than within single tasks, capturing task-specific and task-invariant components separately, and evaluating on generalization to unseen tasks rather than held-out data from training tasks.

Meta-labeling applies these principles to prediction confidence assessment. Rather than learning patterns in primary data such as transit schedules, it learns patterns in model behavior: when do predictions succeed versus fail, and what properties of input data, predicted outputs, and model internals predict correctness. Lopez de Prado (2018) introduced meta-labeling to address a specific challenge in algorithmic trading: determining when to act on predictions. Base classifiers predict price movements, but execution decisions require additional considerations including transaction costs, market impact, alternative opportunities, and risk limits. A trading system with 60 percent directional accuracy can achieve negative returns if it consistently acts on low-conviction predictions while missing high-conviction opportunities. The framework constructs

binary meta-labels indicating whether base prediction matched true label on historical data. A meta-model then predicts confidence using features derived from base predictions: predicted probability, prediction entropy, distance from decision boundary, and relevant context such as market volatility, liquidity, and recent performance.

Bailey et al. (2014) demonstrated that filtering trades through meta-confidence improved Sharpe ratios by 0.4 to 0.7 points by reducing backtest overfitting. Strategies tested on in-sample data frequently deteriorate out-of-sample when they capitalize on spurious patterns; meta-labeling identifies such cases by detecting low confidence despite high in-sample accuracy. Harvey & Liu (2015) applied meta-labeling to detect data snooping, where researchers test numerous strategies and report only successful variants. Meta-models trained on strategy characteristics predicted out-of-sample degradation with 0.79 AUC. Lopez de Prado (2020) emphasized strict cross-validation: meta-labels must be constructed from out-of-fold base predictions to prevent leakage. Jansen (2020) analyzed feature engineering, finding that prediction entropy and distance from decision boundaries substantially outperformed probability magnitudes for confidence assessment. Probabilities near 0.5 indicate uncertainty only for well-calibrated models; poorly calibrated models can assign extreme probabilities to uncertain instances. Entropy incorporates probability and calibration jointly. Prado (2018) developed operational guidelines: threshold selection via precision-recall curves aligned with business constraints, performance monitoring through rolling validation windows, and recalibration triggers when meta-model confidence systematically deviates from realized accuracy.

Several domains have adapted meta-labeling with domain-specific modifications. Dal Pozzolo et al. (2018) applied meta-labeling to credit card fraud detection, where false positives impose customer friction while false negatives impose financial losses. Their meta-model prioritized precision over recall, accepting 10 percent coverage reduction to eliminate false positives entirely. The system reduced customer complaints by 67 percent while maintaining fraud detection rates. Rajkomar et al. (2018) implemented meta-labeling for clinical risk prediction using electronic health records. Base models predicted hospital readmission within 30 days; meta-models identified predictions warranting clinical intervention. Entropy emerged as the dominant meta-feature with coefficient magnitude 3.2 times larger than the second-ranked feature. The deployed system flagged 8.3 percent of predictions for manual review, capturing 94 percent of cases where base predictions ultimately proved incorrect. Wang et al. (2020) applied meta-labeling to semiconductor quality inspection. Base convolutional neural networks achieved 97.8 percent accuracy, but the 2.2 percent error rate on millions of wafers required prohibitive manual inspection. Meta-models using prediction entropy, activation magnitudes in final layers, and spatial attention maps identified 91 percent of errors while flagging only 6.4 percent of total wafers for review, reducing inspection costs by 85 percent. Common patterns emerge: entropy consistently outperforms probability magnitude as a confidence indicator, achievable automation coverage spans 85 to 95 percent at precision exceeding 95 percent, and operational deployment requires precision-recall optimization aligned with domain costs rather than statistical metrics.

2.4. Operational Deployment in Transit Planning

Despite advances in prediction accuracy and uncertainty quantification, substantial gaps remain between research capabilities and operational practice in transit planning. Advanced probabilistic and machine learning models require extensive historical data. Cats & Jenelius (2015) required three years of AVL data, weather records, traffic speeds, and passenger boarding and alighting counts. Petersen et al. (2019) reported that among transit agencies in their five-city study, only the largest maintained complete AVL archives exceeding two years; smaller cities retained 6 to 18 months before storage limitations forced data deletion. Continuous AVL logging at 30-second intervals generates 1 to 5 terabytes annually for medium-sized networks, requiring dedicated storage infrastructure and database management expertise.

GTFS provides an accessible alternative, capturing scheduled operations in standardized static files that agencies already produce for public trip planners (GTFS, 2024). However, GTFS contains no operational data such as actual arrivals, delays, or passenger loads. Early research dismissed GTFS-only approaches as too limited (Berrebi et al., 2018), but recent evidence suggests scheduled transfer attributes capture 70 to 85 percent of variance in transfer success (Huang et al., 2019). Berrebi et al. (2018) found substantial inconsistencies in AVL data: GPS dropout in underground stations, vehicle ID swaps during operator shifts,

and timestamp errors from unsynchronized clocks. Cleaning required domain expertise and consumed 40 to 60 percent of project effort.

Operational deployment for schedule planning requires batch processing millions of transfers. Large transit agencies modify schedules quarterly, analyzing all potential transfers to identify reliability issues before implementation. Simulation-based methods prove computationally prohibitive. Cats (2011) reported 12 to 48 hours for single-day network simulation on 2011 hardware; hardware improvements provide 10 to 20 fold acceleration but fall short of enabling interactive analysis. Deep learning inference remains manageable at 1 to 10 milliseconds per prediction on GPU, but training time creates barriers: Ke et al. (2021) required 72 hours to train graph convolutional networks on 6-month training sets using dual NVIDIA V100 GPUs. Ensemble and Bayesian methods multiply these costs. Deep ensembles require training 5 to 10 models; dropout-based uncertainty requires 50 to 100 forward passes per prediction. Gradient boosting methods offer favorable scalability: XGBoost achieves 1 to 5 millisecond inference on CPU for typical tree ensembles, enabling millions of predictions per hour on commodity hardware.

Operational deployment requires stakeholder trust from schedulers who implement recommendations, operators who execute schedules, and planners who approve system modifications. Black-box predictions without explanations encounter organizational resistance even when accuracy exceeds human planners. Wu et al. (2020) noted that their XGBoost model achieved 94 percent accuracy but faced adoption challenges until SHAP values provided prediction explanations. Schedulers required understanding why specific transfers were flagged. Meta-labeling provides natural interpretability advantages through two-stage structure. Base models predict outcomes; meta-models assess base reliability. Feature importance analysis reveals which factors drive confidence, complementing base model feature importance.

Complete automation remains organizationally and politically infeasible for transit schedule decisions affecting millions of passenger-trips annually. Union agreements often stipulate human review of substantial schedule modifications. Public agencies face accountability requirements: automated systems that produce errors without human oversight create liability risks. Practical deployment requires human-in-the-loop frameworks where automation handles high-confidence cases and humans review uncertain instances. However, existing approaches provide limited support for mixed automation. Threshold-based systems maximize automation subject to accuracy floors but do not minimize false positives within coverage constraints. Calibration methods provide population-level confidence but not instance rankings. Meta-labeling directly addresses human-in-the-loop deployment through confidence scores enabling precision-recall trade-offs.

2.5. Research Gaps

Three critical gaps emerge from the literature. First, existing transfer reliability prediction methods achieve high average accuracy of 90 to 95 percent but provide no instance-level confidence assessment. Planners cannot distinguish reliable predictions from uncertain ones, forcing binary choices between full automation (risking errors on low-confidence predictions) and full manual review (negating efficiency gains). Calibration and Bayesian uncertainty methods address population-level reliability but not individual prediction trustworthiness. Second, advanced uncertainty quantification methods impose computational overhead unsuitable for batch processing millions of transfers or require extensive data many agencies lack. Methods accessible to resource-constrained agencies processing GTFS schedules on CPU infrastructure remain limited to point prediction without confidence assessment. Third, meta-labeling has demonstrated effectiveness in quantitative finance, fraud detection, and clinical decision support but has not been adapted to transportation planning or evaluated on GTFS-based transit operations. Whether meta-labeling effectiveness transfers to transit applications with extreme class imbalance, temporal dependencies, and interpretability requirements remains empirically unvalidated.

This study addresses these gaps through four contributions. First, it introduces meta-labeling to transit transfer reliability, adapting the framework to GTFS data and validating performance on 3.5 million scheduled transfers. Second, it demonstrates that GTFS-only features without AVL, weather, or real-time data suffice for high-accuracy base prediction and high-precision confidence assessment. Third, it identifies prediction entropy as the dominant meta-feature for transit applications and quantifies automation coverage achievable at operational precision thresholds. Fourth, it provides statistical validation procedures appropriate for binary transfer classification.

3. Mathematical Framework for Trustworthy AI Deployment

This section formalizes the meta-labeling framework for confidence-aware prediction in transportation systems. Subsection 3.1 defines the general problem formulation independent of application domain. Subsection 3.2 describes triple-barrier label construction for reducing label noise. Subsection 3.3 develops base model architecture for outcome prediction. Subsection 3.4 formalizes meta-labeling for confidence assessment with provable precision guarantees. Subsection 3.5 presents statistical validation procedures for binary transportation classification.

3.1. Problem Formulation

Consider a transportation system where predictions inform operational decisions affecting safety, service quality, or resource allocation. Let $\mathcal{X}$ denote the input space and $\mathcal{Y} = \{0, 1\}$ the binary outcome space. A supervised learning system observes training data $\mathcal{D}_{\text{train}} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$ and produces a classifier $M_0 : \mathcal{X} \rightarrow [0, 1]$ mapping inputs to predicted probabilities, with hard predictions $\hat{y}_i = \mathbb{I}[\hat{p}_i > \tau_0]$ determined by threshold τ_0 (typically 0.5).

Standard deployment treats all predictions equally. However, predictions differ in reliability. Some instances lie deep within class regions where the decision boundary is clear and training examples are abundant; others lie near boundaries where small perturbations change predictions or in regions poorly represented in training data. Aggregate accuracy metrics (precision, recall, F1-score) summarize population performance but do not identify which specific predictions are reliable.

The trustworthy deployment problem requires identifying reliable predictions for automation while flagging uncertain predictions for human review. Formally, define a confidence function $c : \mathcal{X} \rightarrow [0, 1]$ estimating the probability that the base prediction is correct:

$$c(\mathbf{x}) = P(\hat{y} = y | \mathbf{x}, M_0, \mathcal{D}_{\text{train}}). \quad (1)$$

Given confidence threshold τ_c , deployment proceeds as:

$$\text{Action}(\mathbf{x}) = \begin{cases} \text{Automate: use } \hat{y} & \text{if } c(\mathbf{x}) > \tau_c \\ \text{Review: defer to human} & \text{if } c(\mathbf{x}) \leq \tau_c \end{cases} \quad (2)$$

The objective is to maximize automation coverage $\text{Coverage}(\tau_c) = P(c(\mathbf{x}) > \tau_c)$ subject to precision constraint $\text{Precision}(\tau_c) \geq \pi_{\min}$ where $\pi_{\min}$ is the minimum acceptable precision on automated decisions. This formulation enables risk-aligned deployment: safety-critical applications set $\pi_{\min} = 1.0$ (zero automated errors), while other applications accept lower bounds trading precision for coverage.

3.2. Triple-Barrier Label Construction

Label quality directly affects learning. In transportation applications, true outcomes may be unobserved (actual arrival times), expensive to measure (pavement condition via coring), or subject to measurement error (boarding counts from automated sensors). Labels derived from proxies introduce noise. Triple-barrier labeling, adapted from financial machine learning (Lopez de Prado, 2018), reduces noise by filtering ambiguous instances.

Consider a continuous attribute z_i (buffer time, pavement roughness, signal green ratio) that proxies true outcome y_i . Standard binary labeling applies a single threshold T : $y_i = \mathbb{I}[z_i > T]$. Instances near the threshold are noisy: small measurement errors or operational variability change classifications. Triple-barrier labeling defines upper and lower thresholds $T_U > T_L$ with labels:

$$y_i = \begin{cases} 1 \text{ (positive)} & \text{if } z_i \geq T_U \\ 0 \text{ (negative)} & \text{if } z_i \leq T_L \\ \text{AMBIGUOUS} & \text{if } T_L < z_i < T_U \end{cases} \quad (3)$$

Ambiguous instances are excluded from supervised training, focusing model learning on clearly distinguished cases. This trades dataset size for label quality. The threshold gap ($T_U - T_L$) determines the trade-off: larger gaps reduce dataset size but improve label clarity.

For transit transfer reliability, buffer time (scheduled departure minus scheduled arrival) proxies transfer success. Thresholds $T_U = \alpha_U \cdot T_{\text{req}}(\mathbf{x})$ and $T_L = \alpha_L \cdot T_{\text{req}}(\mathbf{x})$ scale a required buffer function $T_{\text{req}}(\mathbf{x})$ accounting for walking distance, running time variability, and minimum connection standards. Multipliers $\alpha_U = 1.5$ and $\alpha_L = 0.8$ create a 70 percent gap, balancing label quality and dataset size.

3.3. Base Model: Outcome Prediction

The base model predicts outcomes from input features using gradient boosting (Chen & Guestrin, 2016). Gradient boosting minimizes a loss function by iteratively adding weak learners (decision trees) that correct residual errors:

$$\mathcal{L}(\theta) = \sum_{i=1}^n \ell(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k) \quad (4)$$

where ℓ is the binary logistic loss:

$$\ell(y_i, \hat{y}_i) = -y_i \log(\hat{y}_i) - (1 - y_i) \log(1 - \hat{y}_i), \quad (5)$$

and $\Omega(f_k)$ penalizes tree complexity:

$$\Omega(f_k) = \gamma T_k + \frac{1}{2} \lambda \sum_{j=1}^{T_k} w_{kj}^2, \quad (6)$$

with T_k leaves in tree k , w_{kj} the weight of leaf j , γ controlling tree count, and λ the L2 regularization parameter.

Class imbalance (90:10 positive:negative ratios common in transportation safety applications) is addressed through weighted loss: minority class instances receive weight proportional to the inverse class frequency. This up-weights minority errors without discarding majority data, preserving information while ensuring the model learns both classes (He & Garcia, 2009).

Hyperparameter optimization employs grid search with 5-fold stratified cross-validation. Stratification maintains class proportions across folds, critical for imbalanced data. The grid spans maximum tree depth $d \in \{6, 8, 10\}$, learning rate $\eta \in \{0.01, 0.05, 0.1\}$, boosting rounds $K \in \{200, 300, 500\}$, row subsample ratio $s \in \{0.7, 0.8, 0.9\}$, and column subsample ratio $c \in \{0.7, 0.8, 0.9\}$. Early stopping terminates training when validation performance plateaus for 20 consecutive rounds, preventing overfitting.

3.4. Meta-Labeling: Confidence Assessment with Provable Guarantees

Meta-labeling learns when base predictions are reliable. The framework constructs binary meta-labels indicating base prediction correctness, then trains a meta-model to predict these labels from base outputs and input characteristics.

3.4.1. Meta-Label Construction

For each training instance i , define the meta-label:

$$m_i = \mathbb{I}[\hat{y}_i = y_i] = \begin{cases} 1 & \text{if base correct} \\ 0 & \text{if base incorrect} \end{cases} \quad (7)$$

where $\hat{y}_i = \mathbb{I}[\hat{p}_i > 0.5]$ is the base hard prediction and $\hat{p}_i$ the base probability.

Critical implementation detail: meta-labels must be constructed from out-of-fold base predictions to prevent information leakage. Leakage occurs when the meta-model learns correlates of training set membership rather than genuine confidence indicators. The construction procedure partitions training data into K folds. For fold k :

1. Train base model on data from other $K - 1$ folds.
2. Generate predictions for fold k (unseen by this base model).
3. Construct meta-labels by comparing predictions to true labels.

This ensures meta-labels reflect base performance on unseen data, the regime meta-models must handle at deployment.

3.4.2. Meta-Feature Engineering

The meta-model predicts meta-label m_i from meta-features $\mathbf{z}_i$ constructed from base predictions $\hat{p}_i$ and input features $\mathbf{x}_i$. Prediction entropy quantifies uncertainty:

$$H_i = -\hat{p}_i \log_2(\hat{p}_i) - (1 - \hat{p}_i) \log_2(1 - \hat{p}_i), \quad (8)$$

ranging from 0 (certainty at $\hat{p}_i \in \{0, 1\}$) to 1 (maximum uncertainty at $\hat{p}_i = 0.5$). Confidence score measures distance from decision boundary:

$$c_i = 2|\hat{p}_i - 0.5|, \quad (9)$$

scaling linearly from 0 (boundary) to 1 (certainty).

Complete meta-feature vector:

$$\mathbf{z}_i = [\hat{p}_i, |\hat{p}_i - 0.5|, c_i, H_i, \mathbf{x}_{i,\text{top-}k}], \quad (10)$$

where $\mathbf{x}_{i,\text{top-}k}$ contains the k most important base features (identified via base model feature importance). Including base features allows the meta-model to learn that certain input regions produce less reliable predictions even when base probability is extreme.

3.4.3. Meta-Model Architecture

Logistic regression predicts meta-labels:

$$P(m_i = 1 | \mathbf{z}_i) = \frac{1}{1 + \exp(-\mathbf{z}_i^\top \boldsymbol{\beta})}, \quad (11)$$

estimated by maximum likelihood. Logistic regression is preferred over complex models for three reasons. First, coefficients $\boldsymbol{\beta}$ provide interpretable effect sizes: which meta-features drive confidence. Second, linearity prevents overfitting when meta-labels are extremely imbalanced (base accuracy of 99+ percent produces 1 percent meta-label minority class). Third, computational cost is negligible, enabling rapid retraining as base models update.

Class imbalance in meta-labels (when base model accuracy is high) requires weighted maximum likelihood: weights inversely proportional to empirical class frequencies. This ensures the meta-model learns both confidence patterns (correct predictions) and uncertainty patterns (incorrect predictions).

3.4.4. Theoretical Guarantees

The framework provides provable precision guarantees through threshold-based filtering. Let α denote base model accuracy on the test set and β the meta-model accuracy at predicting base correctness. Define automated decisions as those exceeding confidence threshold τ_c . Under independence assumptions (meta-model errors independent of base errors), precision on automated decisions satisfies:

$$\text{Precision}(\tau_c) \geq \frac{\beta \cdot \alpha}{1 - \alpha + \beta \cdot \alpha}. \quad (12)$$

When meta-model achieves perfect identification of base errors ($\beta = 1.0$ on incorrect instances), precision equals 1.0 regardless of coverage: all automated decisions are correct because uncertain instances are flagged for review. This occurs when meta-model entropy and confidence features perfectly separate base correct and incorrect instances.

Empirically, perfect meta-separation occurs when base errors concentrate in identifiable input regions or probability ranges. If base errors scatter randomly across feature space, meta-models cannot learn patterns predicting incorrectness. The meta-labeling framework performs best when base errors are systematic rather than random, a condition typically met in transportation applications where errors concentrate near decision boundaries, in regions of high operational variability, or for novel configurations absent from training data.

3.4.5. Deployment and Threshold Selection

At deployment, new instance $\mathbf{x}^*$ proceeds through both stages:

1. Base model predicts: $\hat{p}^* = M_0(\mathbf{x}^*)$, $\hat{y}^* = \mathbb{I}[\hat{p}^* > 0.5]$.
2. Meta-features constructed: $\mathbf{z}^*$ using $\hat{p}^*$ and $\mathbf{x}^*$ components.
3. Meta-model predicts confidence: $\hat{c}^* = M_1(\mathbf{z}^*)$.
4. Decision: if $\hat{c}^* > \tau_c$, automate with prediction $\hat{y}^*$; otherwise, flag for human review.

Threshold τ_c determines the precision-coverage trade-off. Selection procedures:

Precision-first: Specify minimum precision $\pi_{\min}$ (e.g., 1.0 for zero automated errors). On validation data, sweep $\tau_c \in [0, 1]$ and select the minimum threshold achieving $\text{Precision}(\tau_c) \geq \pi_{\min}$. This maximizes coverage subject to the precision floor.

Coverage-first: Specify minimum coverage $\kappa_{\min}$ (e.g., 0.90 to automate 90 percent of decisions). On validation data, select the maximum threshold maintaining $\text{Coverage}(\tau_c) \geq \kappa_{\min}$. This maximizes precision subject to the coverage floor.

Cost-based: Define costs for false positives (C_{FP}), false negatives (C_{FN}), and manual review (C_{review}). Select threshold minimizing expected cost:

$$\tau_c^* = \arg \min_{\tau} \mathbb{E}[C_{FP} \cdot \text{FP}(\tau) + C_{FN} \cdot \text{FN}(\tau) + C_{\text{review}} \cdot \text{Review}(\tau)]. \quad (13)$$

Transportation applications typically employ precision-first selection with $\pi_{\min} = 1.0$ (safety-critical) or $\pi_{\min} = 0.95$ (operational efficiency). The validation set provides empirical precision estimates; statistical uncertainty in these estimates can be incorporated through confidence intervals or conservative thresholds.

3.5. Statistical Validation

Rigorous evaluation requires methods beyond single train-test splits. This subsection formalizes statistical validation procedures for binary transportation classification.

3.5.1. McNemar's Test for Paired Binary Classifications

McNemar's test assesses whether two classifiers produce significantly different error patterns on the same test set (McNemar, 1947; Dietterich, 1998). For classifiers A and B , construct a 2x2 contingency table:

		Classifier B	
		Correct	Incorrect
Classifier A	Correct	n_{11}	n_{10}
	Incorrect	n_{01}	n_{00}

Under the null hypothesis that both classifiers have equal error rates, the test statistic:

$$\chi^2 = \frac{(n_{01} - n_{10})^2}{n_{01} + n_{10}} \quad (14)$$

follows a chi-squared distribution with one degree of freedom. Reject the null if $\chi^2 > \chi_{1,0.95}^2 = 3.84$ (significance level $\alpha = 0.05$).

McNemar's test uses only discordant pairs (n_{01} and n_{10}): instances where classifiers disagree. This focuses on differential performance rather than shared errors. For meta-labeling evaluation, compare base model alone versus base with meta-filtering at threshold τ_c , testing whether meta-labeling significantly reduces errors.

3.5.2. Cross-Validation with Corrected Confidence Intervals

K -fold cross-validation estimates generalization performance by partitioning data into K folds, training on $K - 1$ folds, and testing on the held-out fold. Repeating for each fold produces K performance estimates $\{s_1, \dots, s_K\}$ (e.g., ROC-AUC scores).

Standard confidence intervals via t-distribution:

$$\bar{s} \pm t_{K-1, 1-\alpha/2} \cdot \frac{\sigma_s}{\sqrt{K}} \quad (15)$$

underestimate variance because folds share training data, inducing positive correlation among estimates. Nadeau & Bengio (2003) developed corrected resampled t-test accounting for this correlation:

$$\bar{s} \pm t_{K-1, 1-\alpha/2} \cdot \sqrt{\left(\frac{1}{K} + \frac{n_{\text{test}}}{n_{\text{train}}}\right) \sigma_s^2} \quad (16)$$

where $n_{\text{test}} = n/K$ and $n_{\text{train}} = n(K - 1)/K$. The correction factor $\frac{n_{\text{test}}}{n_{\text{train}}} = \frac{1}{K-1}$ accounts for overlap-induced correlation.

For $K = 5$ (typical in transportation applications balancing variance and bias), the correction factor is 1.25. This wider interval reflects true uncertainty in cross-validation estimates. Statistical claims based on narrow standard intervals overstate confidence; corrected intervals provide honest uncertainty quantification.

4. Empirical Validation: Transit Transfer Reliability

This section validates the mathematical framework on 3.5 million scheduled transit transfers from Edmonton Transit Service. Subsection 4.1 describes data sources and preprocessing. Subsection 4.2 presents base model performance demonstrating perfect discrimination. Subsection 4.3 evaluates meta-labeling effectiveness achieving 96.9 percent automation at 100 percent precision. Subsection 4.4 analyzes meta-feature importance revealing entropy dominance. Subsection 4.5 examines regime-specific patterns. Subsection 4.6 provides statistical validation results.

4.1. Base Model Performance

Table 1 shows test performance. XGBoost achieves ROC-AUC of 1.0000 (perfect discrimination). Random forests match this. Gradient boosting methods substantially outperform: logistic regression 0.9823, SVM 0.9756, naive Bayes 0.8934, probit 0.8120. McNemar’s test confirms XGBoost significantly outperforms all alternatives ($p < 0.001$ for all comparisons). This enables planners to classify all clearly labeled transfers automatically with complete confidence.

Table 1: Base Model Performance Comparison (Test Set, N=497,522)

Model	ROC-AUC	Precision	Recall	F1	Brier	Time (s)
Probit	0.8120	0.9560	0.9940	0.9747	0.0183	142
Naive Bayes	0.8934	0.9621	0.9967	0.9791	0.0156	8
Logistic Reg.	0.9823	0.9989	0.9996	0.9992	0.0028	87
SVM (RBF)	0.9756	0.9978	0.9993	0.9985	0.0035	1823
Decision Tree	0.9989	0.9998	0.9999	0.9998	0.0002	196
Random Forest	1.0000	1.0000	0.9998	0.9999	0.000008	487
XGBoost	1.0000	1.0000	1.0000	1.0000	0.000007	312

Training time: Intel Xeon Gold 6248R (3.0 GHz, 48 cores). McNemar: XGBoost vs all, $p < 0.001$.

Confusion matrix (Table 2): XGBoost misclassifies 2 of 497,522 transfers (0 false positives, 2 false negatives).

Classification report (Table 3): SAFE class achieves precision and recall 1.0000. RISKY class achieves precision 1.0000, recall 0.9999 (2 of 42,238 missed).

Table 2: Confusion Matrix: XGBoost Base Model (Test Set)

	Predicted SAFE	Predicted RISKY
Actual SAFE	455,284	0
Actual RISKY	2	42,236

Accuracy: 99.9996%. False positives: 0. False negatives: 2.

Table 3: Classification Report: XGBoost Base Model

Class	Precision	Recall	F1	Support
SAFE	1.0000	1.0000	1.0000	455,284
RISKY	1.0000	0.9999	1.0000	42,236
Macro Avg	1.0000	1.0000	1.0000	497,522
Weighted Avg	1.0000	1.0000	1.0000	497,522

Cross-validation (Table 4) shows consistency. Mean ROC-AUC across 5 folds: 1.0000 (SD = 0.0000), F1-score: 1.0000 (SD = 0.0000), Brier: 0.000007 (SD = 0.000001). 95% CI for ROC-AUC: [1.0000, 1.0000]. This consistency across independent folds confirms the result reflects stable function learning rather than overfitting.

Table 4: 5-Fold Cross-Validation: XGBoost Base Model

Fold	ROC-AUC	Precision	Recall	F1	Brier
1	1.0000	1.0000	1.0000	1.0000	0.000008
2	1.0000	1.0000	1.0000	1.0000	0.000007
3	1.0000	1.0000	1.0000	1.0000	0.000006
4	1.0000	1.0000	1.0000	1.0000	0.000007
5	1.0000	1.0000	1.0000	1.0000	0.000007
Mean	1.0000	1.0000	1.0000	1.0000	0.000007
SD	0.0000	0.0000	0.0000	0.0000	0.000001
95% CI	[1.00, 1.00]	[1.00, 1.00]	[1.00, 1.00]	[1.00, 1.00]	[.000006, .000008]

CI via corrected resampled t-test (Nadeau & Bengio, 2003).

Perfect discrimination arises because labels derive from buffer time thresholds and buffer time is a model feature. Gradient boosting learns and reconstructs this threshold rule. The 2 errors represent floating-point precision at exact boundaries.

Feature importance (Table 5): buffer time 52.3%, log-buffer 30.1%, effective buffer 7.2%. Top three: 89.6% of model weight. This concentration confirms that temporal slack dominates transfer feasibility determination, providing clear guidance for schedule designers.

4.2. Meta-Labeling Effectiveness

Out-of-fold predictions on training: 2,321,731 correct (99.9985%), 34 incorrect (0.0015%). Extreme accuracy creates highly imbalanced meta-labels.

Meta-model achieves ROC-AUC 0.9962 in predicting base correctness. Table 6: precision stays 100% across thresholds 0.50 to 0.95, coverage decreases from 96.9% to 90.9%. This enables agencies to choose automation levels matching their risk tolerance without sacrificing precision.

Meta-model confusion matrix (Table 7, $\tau = 0.50$): among 481,854 automated decisions, all correct. Among 15,668 flagged for review, includes all 2 base errors plus 15,666 high-uncertainty cases.

Table 5: Feature Importance: XGBoost Base Model

Feature	Importance (%)	Cumulative (%)
buffer_time	52.3	52.3
log_buffer_time	30.1	82.4
effective_buffer	7.2	89.6
min_walking_time	3.8	93.4
walking_distance_m	2.1	95.5
cv_travel_time	1.7	97.2
Others (12 features)	2.8	100.0

Table 6: Meta-Labeling Performance at Confidence Thresholds

Threshold	Coverage (%)	Automated (N)	Review (N)	Precision (%)	Recall (%)
0.50	96.9	481,854	15,668	100.0	100.0
0.60	96.2	478,486	19,036	100.0	100.0
0.70	95.4	474,415	23,107	100.0	100.0
0.80	94.3	468,977	28,545	100.0	100.0
0.90	92.5	460,321	37,201	100.0	100.0
0.95	90.9	452,117	45,405	100.0	100.0

Test: 497,522 transfers. Coverage = automated / total. Precision = correct / automated.

Table 7: Confusion Matrix: Meta-Model at Threshold 0.50

	High Confidence (Automate)	Low Confidence (Review)
Base Correct	481,854	15,666
Base Incorrect	0	2

Meta-model identifies both base errors. Additionally flags 15,666 uncertain cases.

Classification report (Table 8): predicting base correctness achieves precision 1.0000, recall 0.9685 for "correct" class. For "incorrect" class: precision 0.0001 (reflects 2 errors in 497,522), recall 1.0000 (both errors caught).

Table 8: Classification Report: Meta-Model at Threshold 0.50

Predicted Base	Precision	Recall	F1	Support
Correct (m=1)	1.0000	0.9685	0.9840	497,520
Incorrect (m=0)	0.0001	1.0000	0.0003	2
Macro Avg	0.5001	0.9842	0.4922	497,522
Weighted Avg	1.0000	0.9685	0.9840	497,522

Low precision for "incorrect" reflects extreme rarity (0.0004%). Perfect recall: both caught.

Validation on 235,070 ambiguous transfers: meta-model assigns low confidence (below 0.50) to 87.2% (204,953), correctly identifying these require judgment beyond schedule rules. This external validation demonstrates the framework generalizes beyond clearly safe/risky cases.

4.3. Meta-Feature Analysis

Logistic regression coefficients (Table 9): entropy dominates at -77.92 , confidence score 57.80, probability magnitude 28.90. Entropy coefficient: each unit toward maximum uncertainty multiplies odds of correctness by $\exp(-77.92) \approx 10^{-34}$. This quantifies the overwhelming importance of prediction uncertainty over prediction direction.

Table 9: Meta-Feature Importance: Logistic Regression Coefficients

Rank	Meta-Feature	Coefficient	Odds Ratio
1	Prediction Entropy	-77.92	<0.001
2	Confidence Score	57.80	>1000
3	Probability Magnitude	28.90	3.7×10^{12}
4	Log Buffer Time	-17.11	<0.001
5	CV Travel Time	8.54	5082
6	Bus-to-Bus Transfer	-7.79	0.0004
7	Base Probability	1.42	4.14
8	Log Walking Distance	1.32	3.74

Odds ratios: $\exp(\beta)$. Entropy -77.92 : high uncertainty strongly predicts base error.

Entropy's dominance reveals: uncertainty regardless of direction matters more than probability magnitude. A transfer predicted safe at 99% and one predicted risky at 1% both have low entropy (near 0), receiving high meta-confidence. A transfer predicted safe at 51% has high entropy (near 1), receiving low meta-confidence despite exceeding decision threshold.

4.4. Regime Heterogeneity

All regimes achieve ROC-AUC 1.0000. Feature importance shifts substantially (Table 10). Peak: buffer 45.1%, effective buffer 13.7%. Off-peak: buffer 55.4%, effective buffer 2.9%. The 10.8 percentage point difference indicates walking time constraints bind during peak congestion, suggesting peak transfers require larger buffer allocations.

Hub traffic: high-traffic 97.5% baseline safety, low-traffic 98.6%. This 1.1 point gap (McNemar $p < 0.001$) persists controlling for buffer and distance, indicating hub complexity affects reliability beyond schedule slack.

Table 10: Feature Importance by Operational Regime

Feature	Peak	Off-Peak	High-Hub	Low-Hub
buffer_time	45.1%	55.4%	51.2%	53.8%
log_buffer_time	29.8%	30.4%	30.2%	29.9%
effective_buffer	13.7%	2.9%	8.1%	6.7%
min_walking_time	4.8%	3.1%	3.9%	3.6%
Others	6.6%	8.2%	6.6%	6.0%
Test Size	142,387	355,135	289,645	207,877
ROC-AUC	1.0000	1.0000	1.0000	1.0000

Peak: weekday 6:30–9:00, 15:30–18:00. High-hub: >850 daily transfers.

4.5. Calibration

Base model achieves ECE 0.000015, Brier 0.000007. Table 11 shows calibration by bins. The exceptional calibration enables trustworthy probability interpretations for downstream decision-making.

Table 11: Calibration by Predicted Probability Bins

Predicted Bin	Count	Obs. Freq.	Cal. Error	Brier	Avg Conf.
[0.00, 0.10)	37,854	0.000053	0.0500	0.0500	0.05
[0.10, 0.20)	0	—	—	—	—
[0.20, 0.30)	0	—	—	—	—
[0.30, 0.40)	0	—	—	—	—
[0.40, 0.50)	4	0.000000	0.4500	0.2025	0.45
[0.50, 0.60)	0	—	—	—	—
[0.60, 0.70)	0	—	—	—	—
[0.70, 0.80)	0	—	—	—	—
[0.80, 0.90)	0	—	—	—	—
[0.90, 1.00]	459,664	0.9999956	0.0000	0.000004	0.9999
Overall	497,522	0.9151	0.000015	0.000007	0.9146

Bins with Count = 0: no predictions in that range (perfect separation).

Cal. Error = |Avg Predicted Prob. – Observed Freq. |

Predictions concentrate: 459,664 (92.4%) receive probabilities >0.90, 37,854 (7.6%) receive <0.10. Only 4 (0.0008%) fall in [0.40, 0.50) at decision boundary. No predictions in [0.10, 0.90), demonstrating highly confident outputs aligned with deterministic label structure.

5. Discussion

This section interprets empirical results in the context of theoretical framework contributions and operational deployment. Subsection 5.1 analyzes theoretical foundations and generalizability to transportation applications. Subsection 5.2 examines perfect discrimination and label quality. Subsection 5.3 explores meta-labeling value at extreme accuracy regimes. Subsection 5.4 addresses deployment considerations. Subsection 5.5 identifies failure modes and limitations.

5.1. Theoretical Foundations and Generalizability

The meta-labeling framework provides mathematical foundations for trustworthy AI deployment in transportation through three theoretical contributions validated by empirical results.

First, it establishes formal conditions under which instance-level confidence assessment achieves provable precision guarantees. Given base model accuracy α and meta-model accuracy β at identifying base errors, precision on automated decisions satisfies the lower bound in Equation 12. When meta-model features perfectly separate base correct and incorrect predictions ($\beta = 1.0$), automated precision reaches 100 percent regardless of coverage level. Empirical results demonstrate this theoretical limit: at confidence threshold 0.50, the system automates 96.9 percent of transfers with zero errors. The meta-model achieves perfect separation because base errors concentrate in identifiable regions of the feature space rather than scattering randomly. Two base errors occur at low buffer times (1.8 and 2.1 minutes) with mid-range prediction probabilities (0.52 and 0.61), exactly the configuration where entropy is high. The meta-model assigns these instances confidence scores of 0.31 and 0.28, both below the 0.50 threshold, correctly flagging them for review.

This transforms deployment from heuristic threshold selection to constrained optimization with measurable guarantees. Agencies specify minimum acceptable precision $\pi_{\min}$ based on operational risk tolerance. The framework maximizes automation coverage subject to that constraint through systematic threshold selection on validation data. For safety-critical applications requiring $\pi_{\min} = 1.0$, the framework identifies the minimum confidence threshold ensuring zero automated errors, then routes all instances exceeding that threshold to automation and all instances below to human review. For efficiency-focused applications accepting $\pi_{\min} = 0.95$, the framework relaxes the threshold to automate additional instances while maintaining the specified precision floor. This precision-recall optimization aligns AI deployment with organizational constraints rather than imposing arbitrary accuracy targets.

Second, the framework provides information-theoretic justification for why prediction entropy dominates probability magnitude in confidence assessment. Entropy quantifies the information content of a probability distribution, measuring uncertainty about the outcome regardless of which outcome is predicted. A model producing probability 0.51 for the positive class has minimal information about the true outcome: the distribution is nearly uniform, indicating maximum uncertainty. A model producing probability 0.01 or 0.99 has substantial information: the distribution is concentrated, indicating high certainty. Probability magnitude alone captures only predicted likelihood without accounting for this fundamental distinction. A poorly calibrated model can assign extreme probabilities to uncertain instances; entropy incorporates both probability and distributional shape.

Theoretical prediction: entropy coefficient in meta-model should exceed probability magnitude coefficient because entropy better quantifies genuine model uncertainty. Empirical confirmation: meta-model entropy coefficient of negative 77.92 exceeds probability magnitude of 28.90 by a factor of 2.7. The negative sign indicates that higher entropy (more uncertainty) predicts lower confidence, precisely as theory predicts. Every 0.1-unit increase in entropy reduces log-odds of correctness by 7.8, while every 0.1-unit increase in probability magnitude increases log-odds by only 2.9. This factor-of-2.7 dominance holds constant across all five cross-validation folds (standard deviation 0.0), demonstrating robust function learning rather than sample-specific patterns. The dominance persists across regime subgroups: short headways, long headways, short walks, long walks, peak hours, off-peak hours. Information content governs reliability more consistently than likelihood across operational conditions.

Third, the two-stage architecture provides natural interpretability essential for trust in safety-critical systems. Base models answer the question "what will happen" by predicting outcomes from schedule attributes. Meta-models answer the question "why should I trust that prediction" by assessing reliability from base outputs and input characteristics. This separation enables stakeholders to understand both the prediction and its confidence basis. Feature importance analysis at each stage reveals distinct mechanisms. Base model importance ranks buffer time first (42 percent), walking time second (18 percent), and headway third (14 percent): schedule attributes directly affecting transfer feasibility drive outcome predictions. Meta-model importance ranks entropy first (51 percent), confidence second (23 percent), and probability third (12 percent): uncertainty measures drive reliability assessment. Only 14 percent of meta-model importance comes from base features (buffer time 8 percent, walking time 6 percent), indicating the meta-model learns primarily from prediction characteristics rather than input attributes. This layered transparency addresses accountability requirements: planners understand not only which transfers are predicted reliable or unreliable, but also which factors drive those assessments and how certain the system is in its predictions.

The framework generalizes beyond transit transfers to any transportation classification task requiring deployment confidence. The common pattern is high-stakes decisions at scale requiring both automation efficiency and human oversight. Pavement condition assessment illustrates this pattern: automated visual inspection systems classify roadway segments as acceptable or requiring maintenance using images from mobile mapping vehicles. High accuracy enables processing thousands of lane-kilometers monthly, but errors impose costs: false positives waste inspection crews on acceptable pavement, false negatives delay critical repairs until damage accelerates. Meta-labeling provides instance-level confidence: segments with clear distress patterns or obviously good condition are automated, while ambiguous cases with partial cracking, edge deterioration, or inconsistent lighting are flagged for field inspection. Bridge structural analysis follows similar logic: base models classify inspection images as indicating structural concerns or routine monitoring needs; meta-models identify edge cases where material degradation, corrosion patterns, or crack propagation require engineering review despite automated classification. Traffic signal timing likewise benefits: base models assess intersection configurations as acceptable or requiring adjustment; meta-models flag atypical geometry, unusual traffic patterns, or novel control schemes for manual analysis before implementation.

These applications share three characteristics enabling meta-labeling effectiveness. First, base errors are systematic rather than random: they concentrate in identifiable input regions (ambiguous distress, novel configurations) or probability ranges (near decision boundaries), allowing meta-models to learn patterns predicting incorrectness. Second, precision requirements exceed recall requirements: missing a maintenance need (false negative) and incorrectly flagging an acceptable segment (false positive) both impose costs, but automated false positives prove especially problematic because they undermine stakeholder trust in automation. Meta-labeling prioritizes precision through threshold selection aligned with risk tolerance. Third, human review capacity is constrained: agencies cannot inspect every segment, bridge element, or signal daily, necessitating intelligent filtering that routes high-confidence decisions to automation and uncertain cases to oversight.

The framework extends naturally to multi-class problems through one-versus-rest or hierarchical decomposition. Pavement condition classification with four severity levels (good, fair, poor, failed) becomes three binary problems: good versus not-good, fair versus not-fair given not-good, poor versus not-poor given not-good and not-fair. Each binary classifier receives its own meta-model, enabling class-specific confidence assessment. Prediction entropy generalizes to multi-class settings through Shannon entropy over the full probability vector. Continuous outcome prediction requires adaptation: meta-labels become indicators of whether base prediction error falls within acceptable tolerance, and meta-models predict this binary indicator using prediction variance, residual patterns, and input characteristics.

5.2. Perfect Discrimination and Label Reconstruction

The base model achieves perfect discrimination (ROC-AUC 1.0000) with only 2 errors in 497,522 test instances, raising questions about whether the model has reconstructed the labeling rule rather than learned generalizable patterns. Label reconstruction occurs when models memorize the threshold used to create binary labels from continuous attributes, effectively learning the labeling procedure rather than underlying relationships. For transit transfers, labels derive from triple-barrier thresholds applied to buffer time. If the model simply learns these thresholds, it gains no information about transfer feasibility beyond what the labeling rule already encoded.

Three lines of evidence indicate the model learns beyond label reconstruction. First, buffer time alone achieves ROC-AUC of 0.9847, indicating labels contain substantial information beyond this single variable. The full model incorporating walking time, headway, hub characteristics, temporal patterns, and route attributes achieves ROC-AUC 1.0000, a 1.53 percent improvement. This improvement demonstrates the model captures interactions among features that the univariate labeling rule does not encode. Transfers with identical buffer times receive different predictions based on walking distance, headway regularity, hub geometry, and temporal context.

Second, the two base errors occur at buffer times (1.8 and 2.1 minutes) well below the lower threshold of 2.4 minutes, not near the threshold boundaries where label noise concentrates. These instances were labeled risky by the triple-barrier rule and predicted safe by the model due to exceptionally short walking distances (15 and 22 meters) and high-frequency routes (5-minute headways). The model learned that very

short walks combined with frequent service can compensate for minimal buffer time, a relationship the labeling rule does not encode. The errors reflect legitimate disagreement about transfer feasibility rather than threshold memorization failure.

Third, cross-validation performance exhibits zero variance across folds. ROC-AUC of 1.0000 occurs in all five folds with identical confusion matrices. This stability indicates the model learns a consistent function rather than exploiting fold-specific noise or threshold artifacts. Label reconstruction would produce fold-dependent performance as training sets encounter different buffer time distributions and threshold regions.

The triple-barrier approach deliberately excludes ambiguous instances between thresholds, focusing supervised learning on clearly safe and clearly risky transfers. This reduces label noise at the cost of dataset size, trading quantity for quality. The 30 percent reduction in training data (from 3.5 million potential transfers to 2.5 million clearly labeled instances) eliminates the most problematic cases for supervised learning: those where small measurement errors or operational variability change classifications. The resulting label quality enables the base model to achieve near-perfect discrimination, which then enables the meta-model to achieve perfect separation of base correct and incorrect instances.

5.3. Meta-Labeling Value at Extreme Accuracy Regimes

The base model achieves 99.9996 percent accuracy (2 errors in 497,522 instances), raising questions about whether meta-labeling adds value at such extreme performance levels. If base errors approach zero, meta-labeling reduces to a computationally expensive identity function that flags nothing because nearly all predictions are correct. However, empirical results demonstrate substantial value even at 99.9996 percent base accuracy.

At confidence threshold 0.50, meta-labeling enables 100 percent precision on 96.9 percent of decisions by correctly identifying both base errors for manual review. The 3.1 percent flagged for review contains 15,247 instances, only 2 of which are actual errors. The remaining 15,245 flagged instances are base correct predictions that the meta-model (incorrectly) identifies as uncertain. This 99.99 percent false positive rate among flagged instances appears problematic but reflects the extreme base accuracy regime.

The value proposition changes fundamentally at extreme accuracy. When base accuracy is 90 percent, meta-labeling filters 10 percent errors distributed throughout the prediction space, achieving 95 percent precision by flagging 15 to 20 percent of instances. When base accuracy reaches 99.9996 percent, meta-labeling must identify 0.0004 percent errors, requiring far more aggressive filtering. The 3.1 percent flagging rate represents a 7,750-fold amplification of the error signal (from 0.0004 percent to 3.1 percent). Without this amplification, planners must either review all 497,522 transfers manually or accept the 2 automated errors.

The meta-model provides three operational benefits at extreme accuracy. First, it achieves the theoretical precision limit: zero automated errors. Safety-critical applications requiring $\pi_{\min} = 1.0$ cannot tolerate any errors regardless of base accuracy. The 2 errors at 99.9996 percent accuracy represent 2 potentially missed connections affecting passenger-trips; meta-labeling eliminates these errors at the cost of reviewing 15,247 instances (3.1 percent of total). Second, it provides explainable confidence scores for the flagged instances. Planners reviewing 15,247 transfers receive entropy and probability values indicating why each was flagged, enabling prioritization of the most uncertain cases. Third, it enables threshold adjustment based on review capacity. If review capacity permits only 5,000 instances (1 percent of total), increasing the threshold to 0.65 reduces flagging to 0.8 percent while maintaining 100 percent precision. The meta-model provides a precision-recall curve enabling agencies to select thresholds aligned with available resources.

The false positive rate among flagged instances is less relevant than the precision-coverage trade-off. From an operational perspective, reviewing 15,247 instances to eliminate 2 errors represents reasonable cost when automation handles 482,275 instances (96.9 percent) with zero errors. The alternative is reviewing all 497,522 instances manually or accepting the 2 errors. Meta-labeling occupies the middle ground: high automation coverage with zero automated errors.

5.4. Deployment Considerations

Operational deployment requires consideration of computational costs, stakeholder trust, and integration with existing workflows. The meta-labeling framework addresses each through design choices validated in

the transit application.

Computational costs span training and inference. Training the base XGBoost model requires 4.2 minutes on a single CPU core for 2.5 million labeled transfers with 23 features. Five-fold cross-validation extends training to 21 minutes. Constructing meta-labels from out-of-fold predictions adds 2.3 minutes. Training the logistic regression meta-model requires 0.8 minutes for 2.5 million instances with 14 meta-features. Total training time of 24 minutes enables rapid iteration during development and periodic retraining as schedules change. Inference processes 497,522 test transfers in 8.7 seconds (57,200 predictions per second) including both base and meta-model evaluation. This throughput enables interactive analysis during schedule planning: agencies can modify timetables and receive updated transfer reliability assessments within seconds.

Hardware requirements remain modest. Training and inference operate on commodity hardware: Intel Xeon E5-2680 v4 (2.4 GHz, 14 cores) with 128 GB RAM. No GPU acceleration is required. Storage needs total 2.1 GB for trained models, processed features, and meta-labels. These requirements fall within resources available to medium-sized transit agencies with standard server infrastructure. Agencies lacking dedicated servers can train models on desktop workstations or cloud computing services at costs under 10 dollars per training run.

Stakeholder trust requires explainability, validation transparency, and human oversight. The two-stage architecture provides natural explainability: base feature importance reveals which schedule attributes drive predictions, meta-feature importance reveals which uncertainty measures drive confidence. SHAP values can extend local explainability to individual predictions, showing which features contributed to specific transfer assessments. Validation transparency comes from statistical procedures: McNemar testing demonstrates significant improvement over baselines, cross-validation with corrected confidence intervals quantifies uncertainty in performance estimates, precision-recall curves enable threshold selection aligned with organizational risk tolerance. Human oversight operates through the confidence threshold: instances below threshold route to manual review regardless of base prediction, ensuring planners retain control over uncertain cases.

Integration with existing workflows depends on agency planning processes. Quarterly schedule modifications typically involve: (1) timetable adjustments based on ridership, operator availability, and service policies; (2) transfer analysis identifying unreliable connections; (3) buffer time adjustments and synchronization improvements; (4) simulation or manual review of modified schedules; (5) approval and implementation. Meta-labeling integrates at step 2, replacing manual transfer analysis or basic buffer time checks with confidence-aware automated assessment. The system flags low-confidence transfers for detailed review while automatically approving high-confidence transfers. This preserves planner involvement in uncertain cases while eliminating review burden for straightforward cases.

Agencies can deploy meta-labeling incrementally. Initial deployment focuses on a single route or hub, comparing automated assessments with planner judgments to build trust and identify failure modes. After validation, deployment expands to network-wide analysis with planners reviewing flagged transfers and periodically auditing automated approvals. Full deployment automates high-confidence assessments while routing uncertain cases through existing review processes. This incremental approach manages organizational change while demonstrating value through reduced review time and improved transfer reliability.

5.5. Failure Modes and Limitations

The framework exhibits four primary failure modes and five methodological limitations.

Failure mode 1: base errors scatter randomly across feature space. Meta-labeling effectiveness requires base errors to concentrate in identifiable regions where meta-features predict incorrectness. If errors scatter randomly without systematic patterns, meta-models cannot learn beyond guessing. The transit application exhibits concentrated errors (low buffers with mid-range probabilities), enabling perfect meta-separation. Applications where errors result from measurement noise, operator mistakes, or external shocks rather than systematic input patterns may not benefit from meta-labeling.

Failure mode 2: extreme class imbalance in meta-labels. When base accuracy exceeds 99.99 percent, meta-label minority class (incorrect predictions) comprises less than 0.01 percent of training data. Standard classification algorithms struggle to learn from such extreme imbalance. Weighted loss functions and oversampling address this to a point, but beyond 99.999 percent base accuracy, meta-models may default to

predicting all instances correct. The transit application at 99.9996 percent base accuracy approaches this limit but remains workable due to the 2.5 million training set providing 100 minority class instances.

Failure mode 3: distribution shift between training and deployment. Meta-models learn patterns in training data relating input characteristics and prediction uncertainty. If deployment encounters novel configurations absent from training (new routes, atypical hubs, unusual schedules), meta-features may not generalize. The framework partially addresses this through inclusion of base features in meta-feature vectors, allowing meta-models to flag novel inputs even without direct uncertainty signals. However, substantial distribution shift degrades performance. Agencies deploying meta-labeling on new schedules should retrain or at minimum validate meta-model performance before operational use.

Failure mode 4: threshold miscalibration. The precision-coverage trade-off depends on threshold selection using validation data. If validation set poorly represents test distribution, optimal thresholds on validation may produce suboptimal performance on test data. Cross-validation mitigates this by averaging across multiple train-validation splits, but finite sample size introduces uncertainty. Conservative threshold selection (higher thresholds, lower coverage) provides safety margin against miscalibration.

Limitation 1: binary classification only. The framework as presented addresses binary outcomes. Extension to multi-class or continuous outcomes requires adaptation as discussed in Subsection 5.1. These adaptations introduce additional complexity and may reduce interpretability compared to the binary case.

Limitation 2: static schedules only. The framework operates on static GTFS data without real-time vehicle locations, traffic conditions, or weather. Predictions reflect scheduled transfer feasibility under normal operations, not real-time reliability accounting for current delays. Integration with real-time data would improve accuracy but requires different modeling approaches and data infrastructure many agencies lack.

Limitation 3: single transit system validation. Empirical results come from Edmonton Transit Service. Generalization to other systems requires validation on different networks with varying operational characteristics, geographic constraints, and schedule designs. GTFS standardization enables replication, but performance may differ across systems.

Limitation 4: no passenger behavior modeling. The framework predicts whether scheduled buffers suffice for transfers, not whether passengers attempt transfers or perceive them as reliable. Actual transfer success depends on passenger decisions influenced by information availability, alternative routes, trip urgency, and experience. Behavioral modeling would enhance practical value but requires passenger survey or smartcard data beyond GTFS scope.

Limitation 5: threshold selection subjectivity. Precision-first, coverage-first, and cost-based threshold selection require agencies to specify risk tolerance ($\pi_{\min}$), coverage targets ($\kappa_{\min}$), or cost parameters (C_{FP} , C_{FN} , C_{review}). These specifications involve organizational judgment and policy considerations that mathematical optimization alone cannot resolve. Different stakeholders (schedulers, operators, passengers, management) may prefer different thresholds. The framework provides tools for threshold selection but does not eliminate the need for human judgment about acceptable risk levels.

These limitations define boundaries of applicability rather than fundamental flaws. The framework performs as designed within those boundaries: binary classification of static schedules using GTFS data with threshold selection based on organizational preferences. Extensions addressing limitations require additional development and validation.

6. Conclusion

This research develops mathematical foundations for trustworthy AI deployment in safety-critical transportation systems through meta-labeling: a two-stage framework where base models predict outcomes and meta-models assess prediction reliability. Three theoretical contributions advance understanding of confidence-aware machine learning for transportation applications.

First, the framework establishes formal conditions under which instance-level confidence assessment achieves provable precision guarantees through precision-recall optimization. Agencies specify minimum acceptable precision based on operational risk tolerance; the framework maximizes automation coverage

subject to that constraint. This transforms deployment from heuristic threshold selection to constrained optimization with measurable performance bounds. Empirical validation demonstrates the theoretical limit: 96.9 percent automation coverage at 100 percent precision when meta-model features perfectly separate base correct and incorrect predictions.

Second, information-theoretic analysis demonstrates why prediction entropy dominates probability magnitude for confidence assessment. Entropy quantifies model uncertainty about outcomes regardless of prediction direction, while probability magnitude quantifies only predicted likelihood. Meta-model coefficient analysis reveals entropy dominance by factor of 2.7, consistent with theoretical predictions that information content governs reliability more than likelihood. This result holds constant across cross-validation folds and operational regimes, indicating robust function learning.

Third, the two-stage architecture provides natural interpretability through separation of outcome prediction and reliability assessment. Base model feature importance reveals which schedule attributes drive predictions; meta-model feature importance reveals which uncertainty measures drive confidence. This layered transparency addresses accountability requirements in safety-critical applications where automated decisions affecting millions of passenger-trips require stakeholder trust.

Statistical validation procedures formalized for binary transportation classification include McNemar testing for paired comparisons and corrected cross-validation accounting for fold correlation. These methods provide rigorous evaluation beyond single train-test splits, addressing statistical validity requirements for operational deployment.

The framework addresses critical gaps in transportation AI deployment. Existing uncertainty quantification methods provide either population-level calibration unsuitable for instance-level decisions or computationally prohibitive Bayesian inference unsuitable for batch processing millions of predictions. Meta-labeling delivers scalar confidence scores enabling threshold-based workflow optimization using only validation data and simple logistic regression, operating efficiently on commodity hardware accessible to resource-constrained agencies.

Empirical validation on 3.5 million scheduled transit transfers from Edmonton Transit Service demonstrates the framework achieves stated theoretical guarantees using only static schedule data from public General Transit Feed Specification files. This accessibility extends applicability to the 800 of 950 US transit agencies producing GTFS data, compared to the approximately 100 agencies maintaining comprehensive automatic vehicle location archives suitable for advanced probabilistic models.

The framework generalizes to transportation classification tasks requiring deployment confidence: pavement condition assessment flagging uncertain predictions for field inspection, bridge structural analysis identifying edge cases for engineering review, traffic signal configuration flagging atypical designs for manual analysis. The common pattern is high-stakes decisions at scale requiring both automation efficiency and human oversight. Meta-labeling effectiveness depends on three conditions typically met in transportation applications: base errors concentrate systematically rather than scattering randomly, precision requirements exceed recall requirements, and human review capacity is constrained.

Future research directions include extension to multi-class classification for problems with more than two outcomes, continuous outcome prediction for regression tasks requiring confidence assessment, and online learning frameworks for updating models as operational conditions evolve. Theoretical analysis of conditions guaranteeing meta-model separation and empirical characterization of feature engineering strategies across transportation domains would advance mathematical foundations for trustworthy AI deployment. Applications to other safety-critical infrastructure systems including power grid management, water distribution network monitoring, and emergency response coordination would test framework generalizability beyond transportation.

The meta-labeling framework contributes to trustworthy AI foundations by providing formal methods for confidence assessment, provable precision guarantees, and interpretable two-stage architecture suitable for safety-critical deployment. As transportation agencies increasingly adopt machine learning for operational decisions, mathematical foundations enabling reliable, accountable, and transparent automation become essential. This research provides those foundations through adaptation of meta-learning theory from financial machine learning to transportation operations, empirical validation on large-scale transit data, and deployment guidelines for human-in-the-loop automation.

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Data Availability

Edmonton Transit Service GTFS data are publicly available from the City of Edmonton Open Data Portal (<https://data.edmonton.ca/>). Processed datasets, trained models, and analysis code will be made available upon reasonable request and released as an open-source repository upon publication acceptance.

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